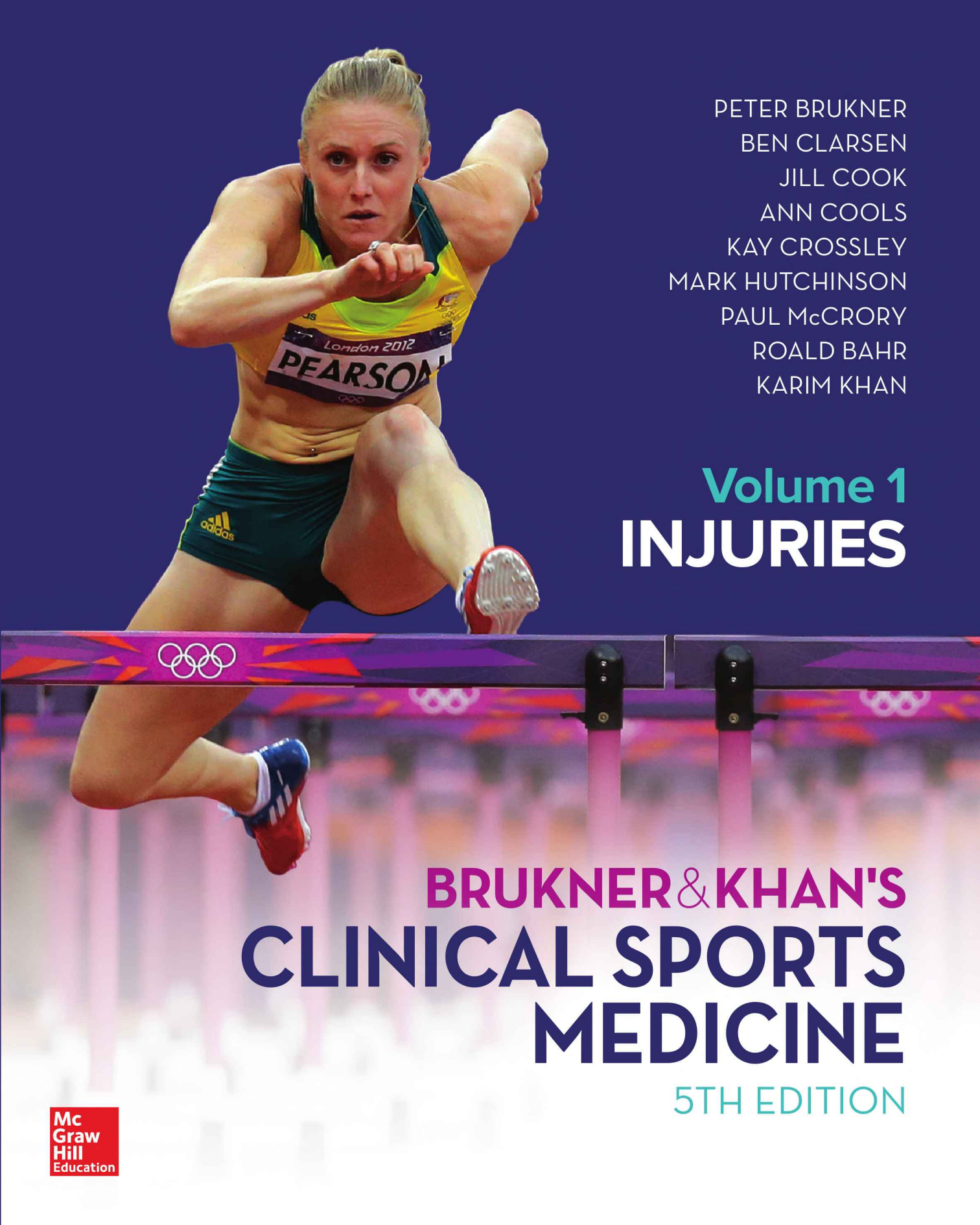




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Volume 1 INJURIES

BRUKNER & KHAN'S CLINICAL SPORTS MEDICINE

5TH EDITION

CLINICAL SPORTS MEDICINE

We dedicate this fifth edition to the Clinical Sports Medicine community—we are proud to be in a family of clinicians who are absolutely unwavering in prioritising patient care.

BRUKNER & KHAN'S

CLINICAL SPORTS MEDICINE

Volume 1
INJURIES

5TH EDITION

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Foreword to the first edition (1993)

Sport in Australia is ingrained in the national consciousness more widely, deeply and indelibly than almost anywhere else in the world. When a prominent sportsperson sustains a sporting injury, either traumatically or from overuse, becomes excessively fatigued, or fails to live up to expectations, this assumes national importance. It is even more relevant nowadays with greater individual participation in sporting activities. The same type of problems occur for recreational athletes, middle-aged people wanting to become fit, or older people wishing to sustain a higher level of activity in their later years.

In *Clinical Sports Medicine* the authors take sport and exercise medicine out of the realm of the elite athlete and place it fairly and squarely where it belongs—as a subspecialty to serve everyone in the community who wishes to be active.

The book is organised in a manner that is sensible and usable. The chapters are arranged according to the anatomical region of the symptom rather than diagnostic categories. This results in a very usable text for the sports physician, general/family practitioner, physiotherapist, masseur, or athletic trainer whose practice contains many active individuals.

Practical aspects of sports medicine are well covered—care of the sporting team and concerns that a clinician might have when travelling with a team. In all, this is an eminently usable text which is timely in its production and will find an important place among clinicians involved in the care of active individuals.

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This foreword was written by the late Professor John Sutton before his untimely death in 1996; we honour the memory of this champion of the integration of science, physical activity promotion and multidisciplinary patient care.

Foreword to the fifth edition

Twenty-four years after the publication of the first edition, the salient point made by Professor John Sutton in his foreword remains true—in *Clinical Sports Medicine* the authors take sports and exercise medicine out of the realm of the elite athlete and present it in the service of everyone in the community who wishes to remain active.

A search for the fourth edition in our university library revealed very well used and even threadbare copies and illustrated how indispensable this resource is to students and practitioners alike. This fifth edition continues the clear focus of previous editions: 'helping clinicians help patients', transcending professional boundaries to enable best practice.

The authors of *Clinical Sports Medicine: Injuries* include the very top names in sports and musculoskeletal physiotherapy, sports medicine and sport science from around the world—in short, the community that takes care of active people everywhere, including the recreational or competitive sportsperson. The truly interdisciplinary nature of professional practice in sports medicine is revealed in this book. Indeed, this fifth edition includes specific chapters on key but often overlooked aspects of professional development; namely working in collaborative teams and how to develop your career. To achieve this, it draws on real-life lessons from around the world.

Clinical Sports Medicine: Injuries builds on the high-quality, evidence-based content of previous editions, with 15 new or substantially revised chapters. It is beautifully illustrated, combining aesthetics and evidence in a partnership that provides accessible learning to all readers. Practice pearls and step-by-step illustrations of physical activity for every joint are supported by more than 550 new and innovative illustrations, and line drawings superimposed on selected photographs to aid understanding.

A striking feature of *Clinical Sports Medicine* has always been the authors' relentless commitment to 'clinical'. This is a useful book; one to be consulted between patients or even while the patient is with you. The colour-coded sections develop, one on another. Part A—Fundamental principles—has been completely rewritten and the cadre of expert authors added new chapters on Pain, Core stability, Patient-reported outcomes (for use in the clinic), Training and Return to sport. Each of the 24 chapters in Part B—Regional problems—stands alone, flowing logically from functional anatomy to the broader, expert clinical perspective on the specific management of more than 100 major conditions. These chapters include the key aspects of injury epidemiology, clinical risk factors and interventions across sports; they cover the common presentations but also provide commentary on the atypical. One aim and notable focus throughout the book is to make evidence accessible to busy practitioners to aid their decision making. Best available evidence, consensus statements and cautionary notes are provided throughout in an eye-catching and succinct way. Part C—Practical sports medicine—closes out Volume 1 with five chapters for those working in various sport settings.

This is a unique book. The variety of illustrations and their contribution to the understanding of form, function, concepts, intervention and management, in combination with decades of the cumulative wisdom of practitioners, teachers and researchers, ensures that this is an indispensable work.

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'Helping clinicians help patients' has been the clear focus of *Clinical Sports Medicine* from its inception. This edition brings you more authors, more artwork and more evidence. We are fortunate that many reading this will already have *Clinical Sports Medicine* in their library, so you are asking, 'What's new?'

- *The complete focus on sports injuries for 1034 pages.* So many advances, we had to make a volume just for injuries. We decided to call it *Injuries*.
- *The giants of sports medicine and sports physiotherapy joining us on the front cover—our seven editors.* You know their work and now they have distilled their best for your benefit (see below and on the Editors page).
- *The artwork.* Vicky Earle's artwork is recognised the world over, the way Dr Frank Netter's was in the 1900s. You may not know the name, but you have seen the tendons, the groin entities and the *BJSM* covers. Five hundred new illustrations to clarify our field.
- *Fifteen new chapters.* (See new content, below.)
- *More than 4000 references—the solid foundation.* As Chapter 2 tells us, we do not yet have all the evidence we would like to have, and clinical advances can outpace the science, but ultimately the scientific method underpins quality care.
- *The unprecedented complementary social media—Facebook, Twitter, Instagram and the www.clinicalsportsmedicine.com website.* This content will have shelf life until 2020 and we will continually update it with the existing digital methods and those that have not yet been invented.

Editors and authors

Volume 1—Injuries reflects the generosity of seven sports physiotherapy and sports medicine editors. Ben Clarsen (senior editor), Jill Cook, Ann Cools, Kay Crossley, Mark Hutchinson, Paul McCrory and Roald Bahr share their passion for helping patients and educating clearly. Our 121 chapter authors and contributors represent 19 countries. Recruited for their first edition, alongside editors Cools and Clarsen, are Rintje Agricola, Hamish Anderson, Sam Blanchard, Jeffrey Boyle, Darren Burgess, Phil Burt, Bojan Buvka, Graeme Close, Natalie Collins, Chad Cook, Adam Culvenor, Michael Davison, Stefano Della Villa, Pieter D'Hooghe, Paul Dijkstra, Sean Docking, Deborah Falla, Rodney French, Tim Gabbett, Phil Glasgow, Adam Gledhill, Alison Grimaldi, Fares Haddad, Toby Hall, Shona Halson, Roger Hawkes, Greg Hoy, Steven Kamper, Vasileios Korakakis, Alex Kountouris, Karl Landorf, Adam Meakins, Håvard Moksnes, Grethe Myklebust, Andrew Nealon, Ian Needlman, Kieran O'Sullivan, Peter O'Sullivan, Jon Patricios, Noel Pollock, Nebojsa Popovic, Michael Rathleff, Ebonie Rio, Ewa Roos, Stephen Simons, Geoffrey St George, Raj Subbu, Stephen Targett, Kristian Thorborg, Elsbeth van Dorssen, Stuart Warden, Adam Weir, Liam West and Fiona Wilson.

New content

Five years is a long time in our field. Major advances, such as high quality evidence for physiotherapy rehabilitation in the management of acute knee injuries, new algorithms for the treatment of shoulder pain and fresh space for the biopsychosocial model are included. The shoulder and knee will remain the shoulder and knee for the next 100 years but chapters can be completely renovated between editions, by new authors. The following chapters have new authors with new diagnostic approaches and new treatments to benefit the clinician.

- Chapter 1: Paul Dijkstra and Stefano Della Villa outline the **team approach** to sports medicine.
- Chapters 3 and 4: Stuart Warden shares the latest mechanistic understanding of **acute** and **overuse injury**.
- Chapter 6: Ebonie Rio builds on Lorimer Moseley's chapter on **pain** (Chapter 5) for the clinician.
- Chapter 9: Experts in their fields share decades each of experience in assessing patients for **sport-specific biomechanical errors**.
- Chapter 10: Darren Burgess gives clinicians insight into the principles that underpin **training** at the very top level of sport and how these apply at the community level.
- Chapter 11: Paul Hodges summarises 20 years of thinking about **core stability**.
- Chapter 14: Chad Cook clarifies the numerical basis that should underpin clinical reasoning and **assessment**.
- Chapter 15: Ali Guermazi partners with fellow radiologist Bruce Forster, and clinician Jon Patricios, to simplify accurate **diagnosis** including MRI interpretation for every clinician.
- Chapter 16: Natalie Collins launches *Clinical Sports Medicine* squarely into the world of PROMs—**patient-reported outcome measures**.
- Chapter 17: Ben Clarsen threw away the fourth edition chapter and built a new **treatment** chapter with international leaders in the different disciplines.
- Chapter 19: Ian Shrier outlines the StARRT model for **return to play**.
- Chapter 21: Toby Hall brings a very practical physiotherapy focus to **headache**.
- Chapter 23: Together with Deborah Falla, Gwen Jull unravels **neck pain** and provides detailed treatment principles and tips.
- Chapter 24: Ann Cools shares her 20 years' experience in assessing and managing **shoulder pain**.
- Chapters 26 and 27: Greg Hoy led the extensive revision of the **wrist pain** and **hand and finger injuries** chapters, highlighting new treatments and new conditions.
- Chapter 29: Peter O'Sullivan incorporates the biopsychosocial approach to **back pain** assessment and management. See the very clinical patient triage outline in Figure 29.2.
- Chapter 30: Adam Meakins describes and illustrates (51 images) how to treat **buttock pain** using exercises as the foundation.
- Chapter 32: Adam Weir skilfully extracted the most useful clinical discoveries from the first world conference on **groin pain**.
- Chapter 35: Sports **knee** surgeons and physiotherapists combine to highlight what is working and what is not working for patients in this rapidly changing field.
- Chapter 43: Karl Landorf and Michael Rathleff are the world leaders on plantar fasciopathy. They provide a major new section inside the important **foot pain** chapter.
- Chapter 46: Stephen Targett applies his vast experience of **periodic medical assessment** to issues such as cardiac clearance, and giving advice that influences contracts worth millions of dollars when doing professional athlete transfer medicals.
- Chapter 47: Liam West hit the refresh button on **working and travelling with teams** and included essentials such as indemnity and emergency preparation.
- Chapter 48: Michael Davison introduces the junior clinician to the global world of sports medicine and sports physiotherapy—A 'how to' on **career development**. (Spoiler alert: hard work is involved!)

No single profession has all the answers required to treat every ill or injured athlete or enthusiast. *Volume 1—Injuries* was created by a champion team. We are confident that, whatever your training, *Clinical Sports Medicine* fifth edition will reinforce and refine existing knowledge and techniques, and introduce useful new approaches for your clinical practice as well as for your teaching of our wonderful vocation.

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**Vicky Earle** BSc (AAM), MET, Cert TBDL

Vicky, an international award winning medical illustrator, has worked with renowned researchers, sports medicine physicians and surgeons around the globe. She has created thousands of illustrations for medical and scientific textbooks and journals, as well as artwork for legal litigation, video, animation and websites on a variety of topics. Vicky's keen interest in *Clinical Sports Medicine* stems not only from a great appreciation of the human body and its capabilities, but also from years of racing experience as a championship rower, a world-class paddler, and a rock climbing enthusiast—and knowing first-hand the many injuries that accompany these activities.

Acknowledgments

Allow me to thank you for being part of this wonderful sports medical book family. I cherish the privilege to have partnered with you to blend science with practice.

This page allows us to say 'thank you'. We open with a quote from one of our chapter authors because it captures the spirit of the large and diverse team that has created Volume 1 of this fifth edition of *Clinical Sports Medicine*. Every single colleague went beyond the call of duty during their busy lives at the peak of great careers.

This highlights that the global sports medicine and sports physiotherapy community is generous and patient focused. Our community shares the wonderful compelling goal of making life better for athletes and for recreational individuals. We share a passion: the joy of physical activity, the love of sport, the conviction that these are elements of medicine at its very best. Here we say 'thank you' to every contributor for allowing us to guide your passion into these pages and the digital elements that complement them.

On the topic of professions in our field, please indulge us a personal note from two medical doctors. To leaders in physiotherapy/physical therapy, we thank you for having embraced the value of physical activity and sport from your profession's beginnings. Physiotherapy/physical therapy has established a very substantial evidence base, along with other disciplines, so that our field–sport and exercise medicine–is recognised as a critical pillar in health. The PEDro database is just one example we can point to if people ask 'What is the scientific basis of the field?' Your profession has been generous to us personally by educating us, and by welcoming us to share our humble opinions with you. We are honoured and grateful to the President of the World Confederation for Physical Therapy for reading this edition and vouching for its value in the foreword and on the back cover. Dr Emma Stokes, we salute you and wish you every success as you execute a shared vision for a very proud, difference-making profession.

Specific thanks for this edition of *Volume 1–Injuries* go to chapter co-authors listed, with their affiliations, on pages xxx–xxxv. We are both humbled and privileged to be sharing cover authorship with seven amazing colleagues and friends: Ben Clarsen, Jill Cook, Ann Cools, Kay Crossley, Mark Hutchinson, Paul McCrory and Roald Bahr. Vicky Earle, our medical illustrator, is part of this book's DNA (three editions now).

The University of British Columbia (Department of Family Practice–Faculty of Medicine, as well as the Faculty of Education–School of Kinesiology) provided essential support (KK), as did La Trobe University in establishing the Sport and Exercise Medicine Research Centre.

We have passed the silver anniversary of McGraw-Hill writing a snail-mail letter to Peter Brukner in 1990 wondering if he thought a sports medicine book might be in order; we are very fortunate for more than 25 years of remarkable support. Nicole Meehan and Robert Ashworth are passionate about this book. Isabella Mead and Yani Silvana always reached out with offers of help. The copy-editing team have added great value for the reader.

We dedicate this fifth edition to the *Clinical Sports Medicine* community. We are proud to be in a family of clinicians who are absolutely unwavering in prioritising patient care. Lastly, to our very, very patient friends and families, 'Yes, it is finished (for now)'. Diana and Heather, thank you for the many editions you have shared us with. Let's leave the conversation about the fifth edition of *Volume 2–Exercise Medicine* (for 2018 release) for another day.

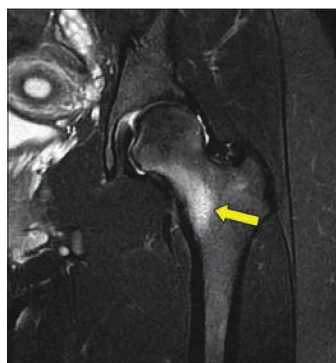
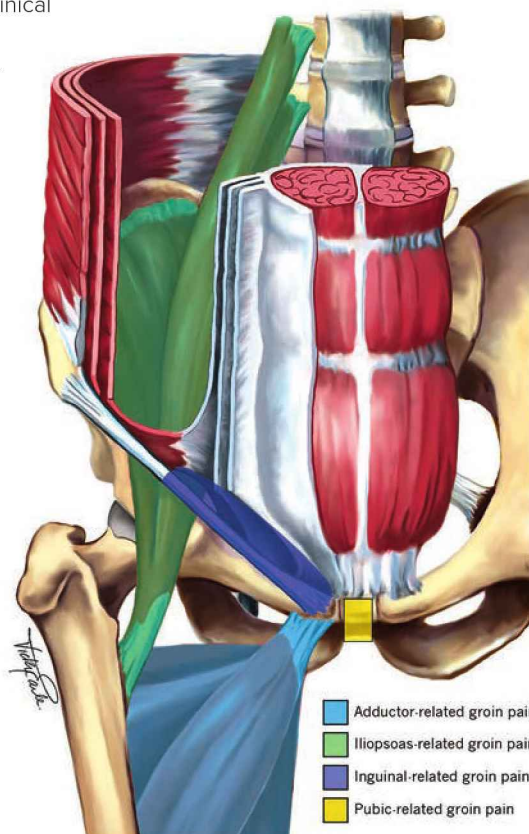
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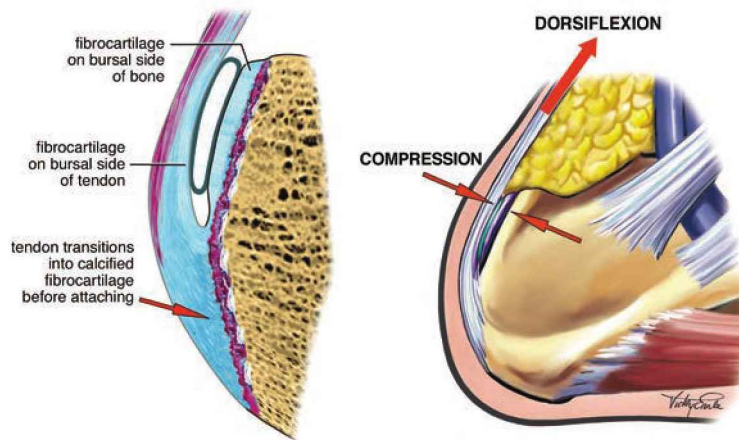
The principal text in its field, this first volume of the fifth edition of *Clinical Sports Medicine* continues to provide readers with quality up-to-date content. The engaging material has been contributed by leading experts from around the world. Look out for these key features, which are designed to enhance your learning.

Premium, up-to-date content

Treatment and rehabilitation is emphasised. Part A—Fundamental Principles includes 12 chapters that are either brand new or written by new authors.

The chapters in Part B, which address regional problems, provide clinical solutions via clear text, abundant clinical photos (e.g. physical examination), relevant imaging and customised medical illustrations.





Tables

New tables summarise vast amounts of evidence to provide take-home messages. Primary sources are readily available.

Table 36.6c Remote factors that can contribute to patellofemoral pain syndrome and their possible mechanisms—increased knee valgus/tibial external rotation

Increased apparent knee valgus/tibial external rotation	Possible contributing factors	Confirmatory assessments
	Structural: • genu varum • tibial varum • coxa varum	Radiographic: • long leg X-ray Clinical assessment: • goniometer/inclinometer
	Inadequate strength/control: • hip external rotators • hip extensors • hip abductors • quadriceps • hamstrings • lumbopelvic muscles	Clinical assessment: • manual muscle test • hand-held dynamometer • biofeedback
	Altered movement patterns	Biomechanical/gait assessment (Chapter 8): • can the patient correct the movement pattern when asked to focus on it?

Practice Pearl

Practice pearls are a valuable feature that provide clinical tips and important information to keep in the forefront of your mind.

PRACTICE PEARL

In the large majority of these hamstring strains, the injured muscle is the biceps femoris (reported as 76–87%). Semimembranosus injury is uncommon; semitendinosus injury is rare.

Contributors

The 121 **world-renowned contributors** bring a truly global perspective to the book.

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References

Over 4000 carefully chosen references. A comprehensive list of references for each chapter can be found here:
www.mhhe.com/au/CSM5e.

